ALICE L WALTON SCHOOL OF MEDICINE

Aortic Regurgitation and Pulmonary Valve Disease

Last updated December 27, 2025 · 18 min read · In Review

Adapted by: Patrick Brooks, Diptiman Bose, Phil Riley

Reviewers: Jordan Finn, Russell Johnson

Adapted from: Aortic Regurgitation

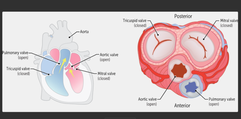
Listen to this Brick

0:00 / 26:15 1x

Learning Objectives (3) Versions

After completing this brick, you will be able to:

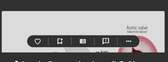
1. Describe the etiologies and pathophysiology of acute and chronic AR
2. Describe the clinical presentation, diagnosis, and management of AR.
3. Discuss Puls… …ologies and aorta. During this event the tricuspid and mitral valves remain closed (Figure 1).



quiz

Figure 1 Blood flow from left ventricle to aorta (left) and cross section showing cardiac skeleton (right)

When the LV relaxes in diastole, the three leaflets of the aortic valve slam together and stop the pressurized blood in the aorta from shooting back into the heart. AR happens when the three aortic leaflets can't close all the way, letting blood flow back from the aorta into the heart during diastole (Figure 2).



Etiologies of Aortic Regurgitation (LO 1)

Causes of Acute Aortic Regurgitation

Acute AR presents with sudden dyspnea, since the LV has no time to compensate for the valvular failure. The main causes are:

- **Acute bacterial endocarditis**: sudden deposition of vegetations on the aortic valve
- **Acute aortic dissection**: separation of the layers of the aortic wall, most common in patients with severe hypertension. If the dissection tracks to the ascending aorta, it can compromise the aortic valve preventing coaptation (Figure 2).
- **Traumatic or surgical injury** can cause regurgitation from any valve.

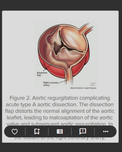


Figure 2. Aortic regurgitation complicating acute type A aortic dissection. The dissection flap distorts the normal alignment of the aortic leaflet, leading to malcoaptation of the aortic…

What do you want to learn?

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Why is penicillin (or erythromycin, if allergic) continued for 5 to 10 years after successful treatment of ARF?

CASE CONNECTION BACK TO INTRODUCTION

Thinking back to BF, how do you explain his condition to him?

An echocardiogram shows thickened mitral valve leaflets and mild mitral stenosis. You explain to BF that his past throat infections caused damage to and narrowing of one of his heart valves. This narrowing is likely to progress, and he may require surgery to replace the valve. You recommend that he see a heart specialist.

Summary

- Acute rheumatic fever (ARF) is a systemic inflammatory disease caused by autoreactive anti… …ees, and lower back. group A *Streptococcus*.
- Rheumatic heart disease (RHD) is a chronic valvular disease that complicates ARF years later.
- ARF occurs in children 1-5 weeks after streptococcal pharyngitis, presenting with fevers, joint and muscle pains, and cardiac findings.
- RHD generally occurs 10-20 years later and leads to chronic valve disease, causing symptoms of heart failure like weakness, dyspnea, and edema.
- ARF may cause carditis, including pericarditis, endocarditis, heart block, mitral valve regurgitation, and aortic regurgitation.
- RHD has characteristic granulomatous lesions called Aschoff bodies that contain Anitschkow cells ("caterpillar" cells).
- RHD comes years after ARF, leading most commonly to mitral or aortic stenosis/regurgitation, and subsequent heart failure.
- Diagnosis of ARF is supported by documented strep infection with carditis, arthritis, chorea, and/or skin nodules.
- Supportive criteria include elevated inflammatory markers, such as erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP).
- Diagnosis of RHD requires an echocardiogram to confirm valvular heart disease.
- Treatment of ARF includes prednisone and penicillin.
- Recurrent pharyngitis and ARF can be prevented with a prolonged course of penicillin.
- RHD with valve stenosis and heart failure may require valve replacement.

Review Questions

1. **A 16-year-old female presents to the clinic with joint pains that began a week ago. The pai… …ees, and lower back.**

What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Acute Rheumatic Fever and Rheumatic Heart Disease

Last updated March 7, 2026 · 17 min read · In Review

Adapted by: Rhorne Hughes, Chelsey Deel, Stephen Nix, Diptiman Bose, Ray Brown, Lori Garman, Patrick Brooks

Editors: Jordan Finn, Russell Johnson

Adapted from: Acute Rheumatic Fever and Rheumatic Heart Disease

Listen to this Brick

0:00 / 28:08 1x

Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Define acute rheumatic fever (ARF) and rheumatic heart disease (RHD).
2. Differentiate the clinical presentation of ARF and RHD.
3. Explain the … …leading to heart failure. Both diseases are common causes of heart disease worldwide but are now uncommon in most developed countries.

Some strains of streptococci can stimulate a strong immune response, often strong enough to attack normal tissue elsewhere in the body (autoimmunity). These two diseases are both immune system complications of strep infections, rather than results of direct infection of the involved organs. Other autoimmune disorders related to strep infection include **scarlet fever** (sandpaper rash) and **postinfectious glomerulonephritis**, a form of kidney disease causing hematuria and kidney failure.

Epidemiology

ARF is a disease of poverty, largely affecting children (80% of cases) in the developing world. It is relatively uncommon in the United States and other developed countries except in focal regions of extreme poverty and limited medical care. The worldwide incidence is 19 cases per 100,000 patients annually, but this rate drops to less than 2 cases per 100,000 patients in developed countries. This difference is likely due to environmental consequences of poverty (e.g., overcrowding, poor sanitation, and limited access to healthcare).

Chronic RHD develops years or decades after ARF, often being diagnosed in patients ages 20 to 40 years. The international prevalence is 100 to 150 cases per 100,000 people, which drops to 1 case per 100,000 in the United States. In contrast, the overall prevalence of heart failure due to all causes in the United States is 800 to 6000 per 100,000 people, depending on age.

When would you expect acute rheumatic fever (ARF) and rheumatic heart disease (RHD) to develop in children who have streptococcal pharyngitis?

What Is the Pathophysiology of Acute Rheumatic Fever and Rheumatic Heart Disease? (LO #3)

ARF is an autoimmune disease in which antibodies react against the heart valves, joints, and brain. This can lead to chronic RHD.

Pathogenesis of Acute Rheumatic Fever

ARF results from an autoimmune reaction by antibodies generated against streptococcal bacteria cross-reacting against cardiac proteins. Specifically, it's a **type 2 hypersensitivity reaction** disease caused by a cross-reactivity between the M protein epitope of *Streptococcus pyogenes* and body self-antigens. *S pyogenes* … …of the larger *Streptococcus* genus (Figure 1).